

Password Strength Analyzer - Project Summary

The Password Strength Analyzer is a Python-based cybersecurity tool designed to evaluate the strength and security level of user passwords using common password security standards.

The tool checks whether a password meets essential security requirements such as:

- Minimum length
- Uppercase letters
- Lowercase letters
- Numbers
- Special characters

The project uses Python Regular Expressions (Regex) to detect password complexity patterns and assign a security rating such as:

- WEAK
- MEDIUM
- STRONG

Features

- Password complexity analysis
- Secure hidden password input using `getpass`
- ASCII banner interface
- Regex-based validation
- Security strength classification
- User-friendly terminal output

Technologies Used

- Python
- Regular Expressions (Regex)
- `getpass` module

Learning Outcomes

This project helps understand:

- Password security principles
- Regex pattern matching
- Input validation techniques
- Cybersecurity automation basics
- Secure coding practices

Possible Future Enhancements

The project can be enhanced by adding:

- Password entropy calculation

- Breached password detection
- Password generator
- GUI interface
- Real-time strength meter
- Security compliance checks

⚠ Disclaimer

This tool is developed for educational and authorized cybersecurity learning purposes only.

🌊 30 Days of Cybersecurity Automation

Building practical cybersecurity tools, one day at a time.

OUTPUT:

```
python password_checker.py

Password
Strength

Password Strength Analyzer

Enter Password:

[+] Running Security Checks ...

[+] Minimum length verified
[+] Uppercase letter detected
[+] Lowercase letter detected
[+] Number detected
[+] Special character detected

Password Strength: STRONG 🔥
```

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```
Enter Password:
```

```
[+] Running Security Checks...
```

```
[+] Minimum length verified
```

```
[+] Uppercase letter detected
```

```
[+] Lowercase letter detected
```

[+] Number detected

```
[-] No special character found
```

11. <http://www.chem.mcgill.ca/~mcmurry/chem353/chem353.htm>

🔒 Password Strength Analyzer

Enter Password:

```
[-] Password too short
[-] No uppercase letter found
[+] Lowercase letter detected
[-] No number found
[-] No special character found
```

Password Strength: WEAK ❌